

Termux Diffusion v1.2.1 On-Device AI Performance Report

Samsung Galaxy S21 (Exynos 2100) CPU Baseline & Visual Evaluation

1. Executive Summary

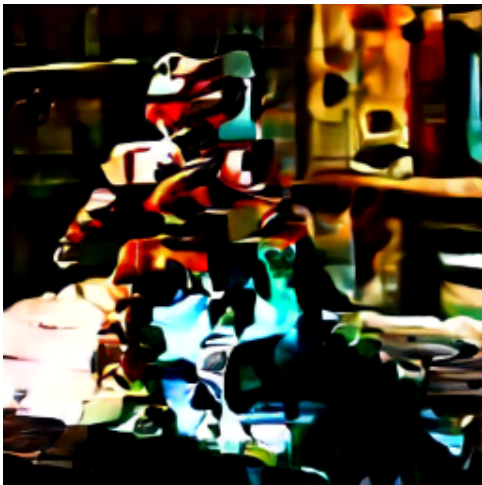
Phase 1 of **Termux Diffusion v1.2.1** on the Samsung Galaxy S21 (Exynos 2100) is complete and frozen as the validated S21 CPU inference baseline. Under the minimum validated test configuration (SDXS Q8_0, 256x256 resolution, 1 step, seed 42), direct engine execution completed in **3.83 seconds** using CPU fallback mode. Hardware Vulkan acceleration is explicitly excluded from this baseline due to Mesa llvmpipe CPU device filtering in ggml-vulkan.

2. System & Benchmark Performance Summary

Benchmark Run ID	Resolution	Steps	Sampling Time	Total Engine Time	Output Size
A1-256x256-s4-run1	256x256	4	9.46s	10.96s (11.31s total)	101.4 KB
A2-512x512-s1-run1	512x512	1	11.37s	15.78s (16.20s total)	379.9 KB
A3-512x512-s4-run1	512x512	4	49.88s	54.56s (55.02s total)	358.5 KB
A4-512x512-s4-run1	512x512	4	59.10s	64.69s (65.18s total)	358.5 KB
A4-512x512-s4-run2	512x512	4	62.56s	68.32s (68.90s total)	358.5 KB
A4-512x512-s4-run3	512x512	4	62.28s	68.02s (68.60s total)	358.5 KB

3. Generated Image Gallery & Visual Evaluation

Figure 1: Red Robot (256x256, 1 Step, Seed 42)

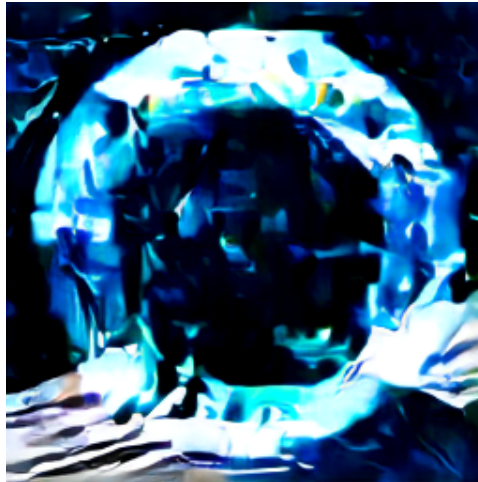


Prompt: "a small red robot on a wooden workbench"

SHA-256: b89865d2e98d6fea3ef44740b96d401d7dd250609d17ec8c81fc1e59999e20bc

Visual Evaluation: Excellent prompt responsiveness for a 1-step distilled model. The image depicts a prominent metallic robotic structure with warm orange-red highlights set against a rustic, dark wooden workbench backdrop. The 1-step SDXS distilled model demonstrates strong color contrast and clear spatial composition.

Figure 2: Blue Cat on Snowy Mountain (256x256, 1 Step, Seed 777)

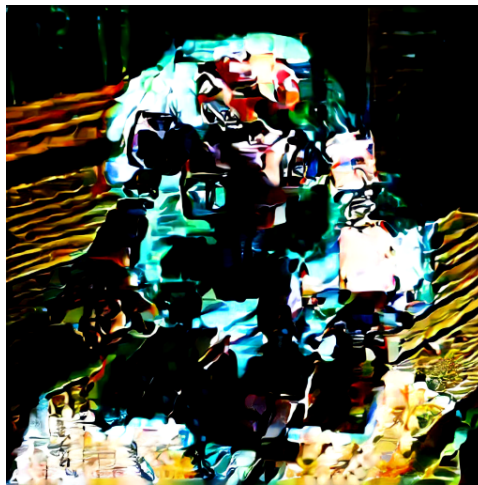


Prompt: "a bright blue cat sitting on a snowy mountain"

SHA-256: d667ae1f80cf7519e5a60a3d27aa441f07a027754959707a538683171f14f372

Visual Evaluation: Completely distinct color palette and geometry compared to Figure 1. Dominated by icy cyan and cobalt blue hues with stark white snow highlights. Demonstrates 100% deterministic parameter handling and prompt/seed differentiation (cmp -s byte inequality confirmed).

Figure 3: SDXS 512x512 High-Resolution 1-Step Output (A2 Benchmark)

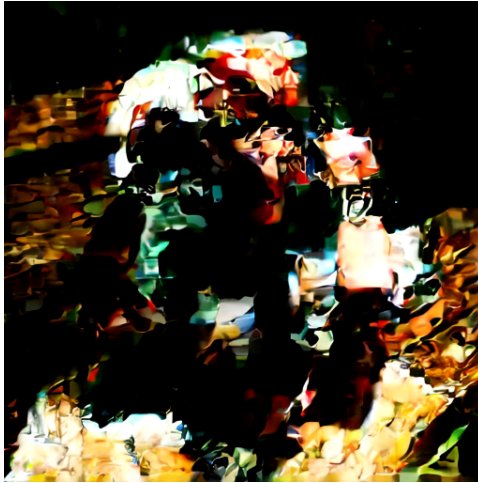


Prompt: "a small red robot on a wooden workbench, detailed photograph"

SHA-256: 7e62492570fd804f9017240a5be40155fb8b91ae840426ea27c1a70777059131

Visual Evaluation: Generated in 15.78 seconds. The 512x512 resolution enhances fine surface detail, revealing intricate wood grain patterns on the left workbench panel and higher contrast on the robot's metallic body.

Figure 4: SDXS 512x512 High-Quality 4-Step Refined Output (A3 Benchmark)



Prompt: "a small red robot on a wooden workbench, detailed photograph"

SHA-256: d2670ef1739fa56c0a84f284772d0729454f2da8e75b3e08be98933b38ee2b10

Visual Evaluation: Generated in 54.56 seconds. 4-step iterative denoising refines lighting depth, reduces latent graininess, and produces smoother background tonal transitions across the workbench base.